# Tier 1 Minimal Model — Basketball Application

## Core Structural Insight

Basketball represents an especially clean application of the framework because:

$$\textbf{selection pool is global} \quad \text{while} \quad \textbf{opportunity pool is local}$$

The NBA draft selects globally across all eligible players, but opportunities to generate draft-relevant signals are allocated locally through: - team membership, - minutes, - role, - usage, - and coaching decisions.

# Core Tier 1 Logic

The minimal proposed mechanism is:

$$\textbf{heterogeneous local pools} + \textbf{finite global selection capacity} + \textbf{local opportunity competition} \rightarrow \textbf{nonli}$$

The goal of Tier 1 is to determine whether these minimal ingredients alone can generate the observed inverted-U relationship.

# Basic Structure

Let player $i$ belong to team $j$ during season $t$.

## Latent Ability

$$A_i$$

where: - $A_i$ = latent player ability.

In empirical applications, the observable own-performance metric (OPM) may serve as a proxy for latent ability $A_i$.

Examples: - basketball: points per minute (PPM), - Army: TB ratio / evaluation quality, - academia: publications, citations, or productivity measures.

Throughout discussion, “OPM” may therefore refer to the domain-specific empirical proxy used to operationalize latent ability.

## Local Pool Quality

Define leave-self-out teammate quality:

$$Q_{jt}^{(-i)} = \frac{1}{|P_{jt}| - 1} \sum_{\ell \in P_{jt}, \ell \neq i} A_\ell$$

where: - $P_{jt}$ = team roster / local opportunity pool, - $Q_{jt}^{(-i)}$ = average teammate quality excluding ego.

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## Observed Signal

Observed signal is generated through both: - individual ability, - and local environmental effects.

$$S_{ijt} = A_i + B(Q_{jt}^{(-i)}) + \varepsilon_{ijt}$$

where: - $S_{ijt}$ = observed player signal, - $B(Q)$ = developmental / signaling benefit from stronger teammates, - $\varepsilon_{ijt}$ = stochastic noise term.

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## Local Opportunity Allocation

Opportunities are allocated locally.

$$O_{ijt} = O(A_i, Q_{jt}^{(-i)}, C_{jt})$$

where: - $O_{ijt}$ = opportunity allocation, - examples include: - minutes, - usage, - touches, - offensive role, - visibility, - $C_{jt}$ = local congestion / competition intensity.

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## Draft-Relevant Realized Signal

Realized draft signal depends on both: - observed ability signal, - and opportunity exposure.

$$R_{ijt} = S_{ijt} \cdot O_{ijt}$$

where: - $R_{ijt}$ = realized draft-relevant signal.

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# Global Selection

Define:

$$\Lambda_t = \text{annual global selection capacity}$$

For the NBA draft:

$$\Lambda_t \approx 60$$

Draft outcome:

$$Y_i = 1$$

if player $i$ is selected among the top $\Lambda_t$ players globally.

Potential probabilistic formulation:

$$P(Y_i = 1) = \text{logit}^{-1}\left(\alpha + \beta_1 A_i + \beta_2 Q_{jt}^{(-i)} - \beta_3 C_{jt}\right)$$

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## Source of the Inverted-U

The central tradeoff:

$$\frac{\partial B(Q)}{\partial Q} > 0$$

but:

$$\frac{\partial O}{\partial Q} < 0$$

at sufficiently high local pool quality.

Interpretation: - stronger environments may improve development and signaling, - but concentrated talent may reduce opportunity share.

Therefore:

$$\frac{\partial P(Y_i = 1)}{\partial Q} = \text{development/signaling gains} - \text{opportunity congestion costs}$$

Potential turning point:

$$B'(Q^*) = G'(Q^*)$$

where: - $Q^*$ = optimal local pool quality, - $G(Q)$ = congestion/opportunity-cost function.

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## Variable Glossary / Model Ontology

| Symbol | Type | Meaning | Formulation / Construction |
|---|---|---|---|
| $i$ | index | player | row-level player identifier |
| $j$ | index | team | team identifier |
| $t$ | index | season | season identifier |
| $A_i$ or $A_{ijt}$ | latent / proxied | player ability or own performance | empirical proxy varies by domain; basketball: $PPM_{ijt}$ |
| $P_{jt}$ | set | local opportunity pool | players on team $j$ in season $t$ |
| $Q_{jt}^{(-i)}$ | constructed variable | leave-self-out local pool quality | $\frac{1}{\lvert P_{jt}\rvert-1}\sum_{\ell\in P_{jt},\ell\neq i} A_{\ell jt}$ |
| $S_{ijt}$ | latent signal | observed/evaluated player signal | $A_i + B(Q_{jt}^{(-i)}) + \varepsilon_{ijt}$ |
| $B(Q)$ | function | developmental / signaling benefit from local environment | increasing benefit term; candidate: $B'(Q) > 0$ |
| $\varepsilon_{ijt}$ | noise term | stochastic signal noise | distribution TBD |
| $O_{ijt}$ | observed / constructed | locally allocated opportunity | minutes, usage, role, touches, visibility |
| $C_{ijt}$ | constructed variable family | local congestion / roster pressure | candidate forms: $C_{ijt}^{\tau}$, $C_{ijt}^{sum}$ |
| $C_{ijt}^{\tau}$ | constructed variable | threshold pressure | $\sum_{\ell\in P_{jt},\ell\neq i} \mathbf{1}(A_{\ell jt} > \tau)$ |
| $C_{ijt}^{sum}$ | constructed variable | total roster pressure / talent mass | $\sum_{\ell\in P_{jt},\ell\neq i} A_{\ell jt}$ |
| $R_{ijt}$ | constructed variable | realized draft-relevant signal | $S_{ijt}\cdot O_{ijt}$ |
| $\Lambda_t$ | parameter | annual global selection capacity | NBA draft: $\Lambda_t \approx 60$ |
| $Y_i$ | outcome | draft outcome | 1 if eventually drafted, else 0 |

## Current Empirical Mapping

Current empirical operationalization:

| Concept | Current Empirical Variable |
|---|---|
| individual ability proxy | points per minute (PPM) |
| local pool quality | leave-self-out teammate PPM |
| success event | eventual NBA draft selection |

## Important Conceptual Clarification

This framework distinguishes:

| Structure | Basketball Example |
|---|---|
| global selection pool | NBA draft |
| local opportunity pool | team roster |
| local opportunity allocation | minutes, role, usage |
| global advancement outcome | draft selection |

This distinction may become one of the central conceptual foundations of the broader theory.

## Addendum — Candidate Definitions of Local Congestion / Roster Pressure

An important emerging distinction is:

$$Q_{jt}^{(-i)} \neq C_{ijt}$$

where: - $Q_{jt}^{(-i)}$ captures average local pool quality, - while $C_{ijt}$ captures local crowding pressure / roster pressure.

This distinction may become theoretically important.

Examples: - a team may have moderate average talent but very high crowding pressure, - or a team may have high average talent concentrated in only one or two players.

Thus: - average environment quality, - and competition for scarce opportunity,

may represent distinct mechanisms.

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## Candidate Interpretation of $C_{ijt}$

Conceptually:

$$C_{ijt} = \text{competition for scarce local opportunity}$$

Examples: - minutes, - usage, - touches, - command opportunities, - authorship prominence, - evaluator attention.

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## Candidate Formulation 1 — Threshold Pressure

$$C_{ijt}^{\tau} = \sum_{\ell \in P_{jt}, \ell \neq i} \mathbf{1}(A_{\ell jt} > \tau)$$

Interpretation: - number of teammates above some meaningful performance threshold.

Potential strengths: - interpretable, - threshold can be tuned empirically, - captures "how many legitimate competitors surround ego?"

Potential limitation: - does not capture intensity above threshold.

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## Candidate Formulation 2 — Total Roster Pressure

$$C_{ijt}^{sum} = \sum_{\ell \in P_{jt}, \ell \neq i} A_{\ell jt}$$

Interpretation: - total local performance pressure generated by surrounding teammates.

Potential strengths: - captures total competitive talent mass, - reflects cumulative crowding pressure, - avoids reducing congestion to simple averages.

Potential limitation: - may require normalization for roster size or opportunity supply.

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## Important Conceptual Distinction

$$Q_{jt}^{(-i)}$$

asks:

> *"How strong is the average local environment?"*

while:

$$C_{ijt}$$

asks:

> *"How much local competition exists for scarce opportunity?"*

These may produce substantially different dynamics.

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## Cross-Domain Interpretation

| Domain | OPM | Local Pool Quality | Crowding / Pressure |
|---|---|---|---|
| Basketball | points per minute (empirical proxy for $A$) | mean teammate quality | count/sum of strong teammates |
| Army | TB ratio / evaluation quality | mean officer quality | concentration of strong officers in SR pool |
| Academia | publications / citations | mean department productivity | concentration of highly productive peers |

Importantly, different congestion formulations may ultimately prove more appropriate in different domains.

The framework should therefore initially treat:

$$C_{ijt}$$

as a family of candidate congestion measures rather than a single finalized variable.

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# Current Direction

The immediate next goal is not full realism, but:

$$\textbf{identifying the minimal mechanism set capable of reproducing the observed empirical shape.}$$

The Tier 1 framework should therefore remain: - modular, - interpretable, - empirically testable, - and computationally simple enough for preliminary simulation and fitting.